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### ABSTRACT

#### WHOSE STANDARD OF CARE? FOREIGN-GUIDELINE AND OUTDATED-VERSION DEVIATIONS OF LLMS FROM VIETNAMESE MINISTRY OF HEALTH GUIDELINES

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**BACKGROUND:** Large language models (LLMs) may give drug doses or diagnostic thresholds from foreign or superseded guidelines instead of current Vietnamese Ministry of Health (MoH) guidelines; because such values exist in other guidelines, learners and clinicians may not notice. The extent and sources of these deviations are unclear.

**OBJECTIVE:** To pilot a method that measures how often an LLM gives the current MoH value, traces wrong answers to their sources, benchmarked against a chance-match level, and compares question languages and prompting conditions.

**METHODS:** Pilot design: we analysed 61 purposively selected recommendations (of 65) from 12 MoH documents in force, quoted verbatim with page numbers. Three groups: conflict (20; differing from at least one named foreign guideline; 15 of these had a decoy, a pre-specified false value for estimating chance matches), version (20; each with an older MoH version) and concordant (the remaining 21; matching foreign guidelines). Qwen3-8B (a small open model, four-bit quantised, run on a laptop) gave short answers in Vietnamese and English under three conditions: no country cue, an MoH cue and the relevant MoH passage. Answers were graded by pre-specified rules; only an AI (Claude) checked the quotations and all 390 short answers (two blinded passes with an AI referee, used for a sensitivity analysis). Statistics: Clopper–Pearson 95%



confidence intervals (CIs); exact McNemar tests pairing the MoH-cue and passage conditions (exploratory; added post hoc, after the sealed outputs were opened).

**RESULTS:** In Vietnamese with the MoH cue, 7/20 conflict-group answers were correct (95% CI 15.4%–59.2%); 2 matched a foreign value, 7 matched no recorded source, and in 4 the rule grader could not read a value (sensitivity analysis with AI-adjudicated labels: 8/20 correct). With the same cue, 9/21 concordant answers were correct and 3/20 version-group answers matched a superseded MoH value. Among the 15 recommendations with a decoy, 2 answers matched a foreign value and 0 matched the decoy. In the conflict group in Vietnamese, 2/20 answers were correct without a country cue and 13/20 with the passage; versus the MoH cue, 9 changed from wrong to correct and 3 from correct to wrong (exploratory McNemar  $p = 0.15$ ). In the same group in English, the three conditions gave 3/20, 5/20 and 13/20 correct answers (exploratory McNemar  $p = 0.02$ ).

**CONCLUSION:** Most wrong answers matched no recorded source, and foreign-guideline matches were not shown to exceed the chance (decoy) level. With the MoH cue, the concordant group also had fewer than half of its answers correct. With the MoH passage, more answers were correct but errors remained; learners and clinicians should check any value an LLM states against the current MoH guideline. Limitations: small purposive sample, one model, AI-only checking, no clinician review, many decoys rated implausible by AI; a correct answer may also coincide with a foreign value. Next: a main study, to be pre-registered, on the current MoH guideline corpus with several open models.

**KEYWORDS:** large language models; clinical practice guidelines; ministry of health; artificial intelligence; model evaluation

**PRESENTATION FORMAT:** Oral

**TOPIC:** AI and Digital Transformation in Healthcare and Medical Education

**NOTE:** This work has not been published in any national or international journal.